

Hatem Ayman

hatem.ayman.508@gmail.com | 01096535619 | Sharkia, Egypt
github.com/hatemabd-elkarim | linkedin.com/in/hatemabd-elkarim

Education

Zagazig University

Oct 2023 – 2028 (Expected)

B.Sc. in Computers and Systems Engineering

GPA: 3.9/4.0

Related Coursework: Object Oriented Programming, Data Structures, Algorithms, Software Engineering, Database Systems, Computer Networks, Operating Systems, Computer Architecture, Discrete Mathematics, Logic Design

Skills

Programming Languages: C, C++, Java, Python, JavaScript, PHP

Backend: RESTful APIs, MVC, Data Validation, Session/Auth Handling, Testing

Libraries/Frameworks: Express JS, Laravel, Flask, JavaFX

Tools/Platforms: Git, GitHub, Linux, VS Code, Postman, Composer, Firebase, Markdown

Databases: MySQL, SQL Server, SQLite

Soft Skills: Teamwork, Time Management, Project Coordination, Leadership, Self-Learning, Problem Solving, Presentation

Experience

Trainee | National Telecommunication Institute (NTI)

Aug 2026 – Sep 2026

- Completed 3 weeks of technical training sessions in full-stack PHP web development, culminating in a team final project.
- Attended 1 week of soft skills training covering freelancing and customer service fundamentals.

Projects

Shoryan | github.com/Shoryan-org

Laravel, MySQL, Firebase

- Full-stack blood donation mobile app connecting donors with patients in real time through location-based matching, automated eligibility reminders, and an AI-powered donor recommendation system. Built a Laravel REST API with token-based authentication, Firebase push notifications, and a FastAPI microservice layer for a RAG-based medical chatbot and a trained donor-matching model.
- Designed and built 25+ RESTful API endpoints covering authentication and core business logic, secured with Laravel Sanctum token-based authentication.
- Implemented a Firebase Cloud Messaging notification service for real-time push alerts on request acceptance, fulfillment, and donor eligibility.
- Automated donor eligibility reminders using Laravel's task scheduler, running as a recurring background job.
- Architected a two-phase OTP email verification system for registration, using Laravel's cache layer for secure, time-limited, hashed OTP storage.

Gryffindor Platform | github.com/Ahmedmhmud/gryffindor-platform

PHP, PDO, Composer, MySQL

- Built a full custom PHP MVC framework and web application from the ground up — without relying on Laravel, Symfony, or any third-party framework — to power a Harry Potter-themed school management platform for student registration, course/quiz management, gamified house points, and an in-app marketplace.
- Architecture:** Custom MVC framework (routing, DI container, session management, PDO-based DB abstraction), fully PSR-4 autoloading; centralized request lifecycle with global exception handling through a single entry point; reusable server-side validation engine with flash messaging for error/old-input persistence.
- Auth & Security:** bcrypt password hashing and session regeneration against fixation; role-based access control via custom middleware (Student/Professor/Admin guards); parameterized PDO prepared statements platform-wide; DB transactions for atomic user and role record creation.

Down-Up | github.com/hatemabd-elkarim/Markdown-to-HTML-converting-tool

Python, Regex

- A Markdown to HTML converting CLI tool that reads any Markdown file, applies rule-based parsing using regex, and outputs a formatted HTML file — useful for docs and static pages.

Sudoku | github.com/hatemabd-elkarim/Sudoku-game

Java, JavaFX

- Desktop Sudoku game in Java with a JavaFX GUI; features 3 difficulty levels, mistake tracking, and a live timer.

Extracurricular Activities

IEEE ZSB – CS Backend Member

Nov 2025 – Aug 2028

- Contributed to 3 team projects throughout the track, including database schema design, a full-stack web application, and a deployed Laravel API.
- Built 20+ PHP tasks covering MySQL schema design, MVC architecture, OOP, and web security, each documented in Markdown.
- Studied common web vulnerabilities (SQL injection, XSS) and how to prevent them.
- Ranked top 3 out of 20+ attendees throughout the track.

ICPC ZSB – Level 1 Trainee

Mar 2026 – May 2026

- Solved 80+ medium-difficulty problems and participated in 5+ contests throughout the training.
- Secured 5th place out of 30+ trainees in the Level 1 (Intermediate) division.